

**MATH3026/4022: Time Series Analysis/Time Series and Forecasting**  
**Exercises Sheet 2 - SOLUTIONS**

1. Suppose that  $\{Z_t\}$  is a white noise process with variance  $\sigma_z^2$ . For each of the following processes,  $\{X_t\}$ , answer the following: (a) Is the process stationary? (b) If the process is stationary, what is its mean? (c) If the process is stationary, what is its autocovariance function?

(i)  $X_t = \frac{5}{4}X_{t-1} - \frac{3}{8}X_{t-2} + Z_t$ .

**SOLUTION:**

(a) Since this is an AR process, we write the process as

$$\phi(B)X_t = Z_t$$

and check whether or not the roots of  $\phi(z) = 0$  lie outside the unit circle.

$$\begin{aligned} X_t - \frac{5}{4}X_{t-1} + \frac{3}{8}X_{t-2} &= Z_t \\ \implies \left(1 - \frac{5}{4}B + \frac{3}{8}B^2\right)X_t &= Z_t. \end{aligned}$$

Then set

$$\begin{aligned} 1 - \frac{5}{4}z + \frac{3}{8}z^2 &= 0 \\ \implies \left(1 - \frac{3}{4}z\right)\left(1 - \frac{1}{2}z\right) &= 0 \end{aligned}$$

which has roots  $4/3$  and  $2$  and each root lies outside the unit circle. Hence the process is (weakly) stationary.

(b) From lectures, we know that the (weakly) stationary  $AR(p)$  process may be written in the form

$$X_t = Z_t + \sum_{j=1}^p \psi_j Z_{t-j}$$

for suitable coefficients  $\psi_j$  that satisfy stationarity conditions. As a result, we see that

$$\mathbb{E}(X_t) = \mathbb{E}(Z_t) + \sum_{j=1}^p \psi_j \mathbb{E}(Z_{t-j}) = 0.$$

(c) Consider

$$X_t = \frac{5}{4}X_{t-1} - \frac{3}{8}X_{t-2} + Z_t \quad (1)$$

Multiplying (1) by  $Z_t$  and taking expectations we obtain

$$\begin{aligned} \mathbb{E}(X_t Z_t) &= \frac{5}{4}\mathbb{E}(X_{t-1} Z_t) - \frac{3}{8}\mathbb{E}(X_{t-2} Z_t) + \mathbb{E}(Z_t^2) \\ &= \mathbb{E}(Z_t^2) \quad (\text{owing to the causality assumption}) \\ &= \sigma_z^2 \end{aligned}$$

Multiplying both sides of (1) by  $X_t$  and taking expectations we obtain

$$\begin{aligned} \mathbb{E}(X_t^2) &= \frac{5}{4}\mathbb{E}(X_{t-1} X_t) - \frac{3}{8}\mathbb{E}(X_{t-2} X_t) + \mathbb{E}(X_t Z_t) \\ \implies \gamma(0) &= \frac{5}{4}\gamma(1) - \frac{3}{8}\gamma(2) + \sigma_z^2 \end{aligned} \quad (2)$$

Next we multiply (1) by  $X_{t-1}$  and take expectations to obtain

$$\begin{aligned} \gamma(1) &= \frac{5}{4}\gamma(0) - \frac{3}{8}\gamma(1) \\ \implies \frac{11}{8}\gamma(1) &= \frac{5}{4}\gamma(0) \\ \implies \gamma(1) &= \frac{10}{11}\gamma(0). \end{aligned} \quad (3)$$

Multiplying (1) by  $X_{t-2}$  and taking expectations yields

$$\begin{aligned} \mathbb{E}(X_t X_{t-2}) &= \frac{5}{4}\mathbb{E}(X_{t-1} X_{t-2}) - \frac{3}{8}\mathbb{E}(X_{t-2}^2) + \mathbb{E}(X_{t-2} Z_t) \\ \implies \gamma(2) &= \frac{5}{4}\gamma(1) - \frac{3}{8}\gamma(0). \end{aligned} \quad (4)$$

Substitution of (3) into (4) yields

$$\begin{aligned} \gamma(2) &= \frac{5}{4} \left( \frac{10}{11}\gamma(0) \right) - \frac{3}{8}\gamma(0) \\ &= \frac{67}{88}\gamma(0). \end{aligned} \quad (5)$$

Now substitute (3) and (5) into (2) to obtain

$$\begin{aligned} \gamma(0) &= \frac{5}{4} \left( \frac{10}{11}\gamma(0) \right) - \frac{3}{8} \left( \frac{67}{88}\gamma(0) \right) + \sigma_z^2 \\ &= \left( \frac{800 - 201}{8 \times 88} \right) \gamma(0) + \sigma_z^2 \implies \left( \frac{704 - 800 + 201}{704} \right) \gamma(0) = \sigma_z^2 \\ \implies \frac{105}{704}\gamma(0) &= \sigma_z^2 \\ \implies \gamma(0) &= \frac{704}{105}\sigma_z^2 \end{aligned}$$

What about  $|h| > 2$ ? Multiplying (1) by  $X_{t-h}$  ( $h > 0$ ) and taking expectations yields the difference equation

$$\begin{aligned}\gamma(h) &= \frac{5}{4}\gamma(h-1) - \frac{3}{8}\gamma(h-2) \\ \gamma(h) - \frac{5}{4}\gamma(h-1) + \frac{3}{8}\gamma(h-2) &= 0.\end{aligned}$$

The auxiliary equation is

$$\begin{aligned}\lambda^2 - \frac{5}{4}\lambda + \frac{3}{8} &= 0 \\ \implies \left(\lambda - \frac{1}{2}\right)\left(\lambda - \frac{3}{4}\right) &= 0\end{aligned}$$

with roots  $\lambda = \frac{1}{2}$  and  $\lambda = \frac{3}{4}$ . Hence the general solution is of the form

$$\gamma(h) = C_1 \left(\frac{1}{2}\right)^{|h|} + C_2 \left(\frac{3}{4}\right)^{|h|}$$

for constants  $C_1$  and  $C_2$ .

For  $h = 0$ :

$$\gamma(0) = C_1 + C_2 = \frac{704}{105}\sigma_z^2$$

For  $h = 1$ :

$$\begin{aligned}\gamma(1) &= \frac{1}{2}C_1 + \frac{3}{4}C_2 = \frac{10}{11}\gamma(0) \\ \implies \frac{1}{2}C_1 + \frac{3}{4}C_2 &= \frac{10}{11} \times \frac{704}{105}\sigma_z^2 \\ &= \frac{640}{105}\sigma_z^2\end{aligned}$$

Solving the above for  $C_1$  and  $C_2$  we obtain

$$C_1 = -\frac{448}{105}\sigma_z^2; \quad C_2 = \frac{1152}{105}\sigma_z^2$$

and

$$\gamma(h) = \frac{\sigma_z^2}{105} \left( 1152 \left(\frac{3}{4}\right)^{|h|} - 448 \left(\frac{1}{2}\right)^{|h|} \right).$$

(ii)  $X_t = \frac{5}{4}X_{t-1} + \frac{1}{8}X_{t-2} - \frac{3}{8}X_{t-3} + Z_t.$

**SOLUTION:**

This is an  $AR(3)$  process, which we write as

$$\left(1 - \frac{5}{4}B - \frac{1}{8}B^2 + \frac{3}{8}B^3\right)X_t = Z_t.$$

The process is (weakly) stationary if the roots of

$$1 - \frac{5}{4}z - \frac{1}{8}z^2 + \frac{3}{8}z^3 = 0$$

lie outside the unit circle.

By inspection,  $z = 1$  is a root of the above since

$$1 - \frac{5}{4} - \frac{1}{8} + \frac{3}{8} = 0$$

and this root does not lie outside the unit circle. Hence the process is not stationary.

(iii)  $X_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}$  where  $\psi_j = (1+j)^{-1}$ .

**SOLUTION:**

We have

$$X_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}$$

and the process is (weakly) stationary if

$$\sum_{j=0}^{\infty} \psi_j^2$$

is finite. Here

$$\sum_{j=0}^{\infty} \psi_j^2 = \sum_{j=0}^{\infty} \frac{1}{(1+j)^2}$$

which is finite. Therefore, this process is (weakly) stationary.

(b) Clearly, the mean of the process is zero because  $\mathbb{E}(Z_t) = 0$  for all  $t$ .

(c) Consider  $h > 0$ . Using results on the general linear process (from Lecture 3) we know that

$$\gamma(h) = \sigma_z^2 \sum_{j=0}^{\infty} \psi_j \psi_{j+h}.$$

Then

$$\begin{aligned} \gamma(h) &= \sigma_z^2 \sum_{j=0}^{\infty} \frac{1}{(1+j)} \times \frac{1}{(1+j+h)} \\ &= \sigma_z^2 \sum_{j=0}^{\infty} \frac{1}{(1+j)(1+j+h)} \end{aligned}$$

Using partial fractions

$$\begin{aligned}
\frac{1}{(1+j)(1+j+h)} &= \frac{A}{1+j} + \frac{C}{1+j+h} \\
\implies 1 &= A(1+j+h) + C(1+j) \\
&= A(1+h) + C + (A+C)j.
\end{aligned}$$

Equating coefficients:

$$\begin{aligned}
1 &= A(1+h) + C \\
0 &= A + C
\end{aligned}$$

Then  $C = -A$  and

$$1 = A(1+h) - A \implies A = \frac{1}{h} \text{ and } C = -\frac{1}{h}.$$

Then

$$\begin{aligned}
\gamma(h) &= \frac{\sigma_z^2}{h} \sum_{j=0}^{\infty} \left[ \frac{1}{j+1} - \frac{1}{1+j+h} \right] \\
&= \frac{\sigma_z^2}{h} \left\{ \sum_{j=0}^{\infty} \frac{1}{j+1} - \sum_{k=h}^{\infty} \frac{1}{1+k} \right\} \quad (\text{setting } k = j+h \text{ in the second sum}) \\
&= \frac{\sigma_z^2}{h} \sum_{j=0}^{h-1} \frac{1}{j+1}.
\end{aligned}$$

Since the process is (weakly) stationary, it follows that, for  $|h| > 1$ :

$$\gamma(h) = \frac{\sigma_z^2}{|h|} \sum_{j=0}^{|h|-1} \frac{1}{j+1}.$$

For  $h = 0$ :

$$\begin{aligned}
\gamma(0) &= \sigma_z^2 \sum_{j=0}^{\infty} \frac{1}{(j+1)^2} \\
&= \sigma_z^2 \sum_{k=1}^{\infty} \frac{1}{k^2} \\
&= \frac{\sigma_z^2 \pi^2}{6} \quad \text{since } \sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}.
\end{aligned}$$

2. The process  $\{X_t\}$  is a mean zero  $AR(2)$  process such that

$$X_t - (\gamma_1 + \gamma_2)X_{t-1} + \gamma_1\gamma_2X_{t-2} = Z_t$$

where  $|\gamma_1| < 1$ ,  $|\gamma_2| < 1$  and  $\{Z_t\}$  is a white noise process with variance  $\sigma_z^2$ .

(a) Explain why  $\{X_t\}$  is a stationary process.

**SOLUTION:**

We write the process as

$$\begin{aligned}(1 - (\gamma_1 + \gamma_2)B + \gamma_1\gamma_2B^2) X_t &= Z_t \\ \implies \phi(B)X_t &= Z_t\end{aligned}$$

where

$$\phi(B) = 1 - (\gamma_1 + \gamma_2)B + \gamma_1\gamma_2B^2.$$

The roots of  $\phi(z) = 0$  are determined by solving

$$\begin{aligned}1 - (\gamma_1 + \gamma_2)z + \gamma_1\gamma_2z^2 &= 0 \\ \implies (1 - \gamma_1z)(1 - \gamma_2z) &= 0.\end{aligned}$$

Hence the roots are  $z = \frac{1}{\gamma_1}$  and  $z = \frac{1}{\gamma_2}$ . Each root lies outside the unit circle because  $|\gamma_1| < 1$  and  $|\gamma_2| < 1$  and it follows that the process is stationary.

(b) Show that  $\{X_t\}$  can be written as an  $MA(\infty)$  process, with the form

$$X_t = \frac{1}{\gamma_2 - \gamma_1} \sum_{k=0}^{\infty} (\gamma_2^{k+1} - \gamma_1^{k+1}) Z_{t-k}.$$

**SOLUTION:**

Clearly

$$\begin{aligned}X_t &= (1 - \gamma_1 B)^{-1} (1 - \gamma_2 B)^{-1} Z_t \\ &= \left( \frac{1}{1 - \gamma_1 B} \times \frac{1}{1 - \gamma_2 B} \right) Z_t \\ &= \left( \frac{A}{1 - \gamma_1 B} + \frac{C}{1 - \gamma_2 B} \right) Z_t \quad (\text{using partial fractions})\end{aligned}$$

We find that

$$A = \frac{-\gamma_1}{\gamma_2 - \gamma_1}; \quad C = \frac{\gamma_2}{\gamma_2 - \gamma_1}.$$

Hence

$$X_t = \frac{1}{\gamma_2 - \gamma_1} \left( \frac{\gamma_2}{1 - \gamma_2 B} - \frac{\gamma_1}{1 - \gamma_1 B} \right) Z_t.$$

Using Taylor expansions

$$\begin{aligned}(1 - \gamma_1 B)^{-1} &= 1 + \gamma_1 B + \gamma_1^2 B^2 + \gamma_1^3 B^3 + \dots; \\ (1 - \gamma_2 B)^{-1} &= 1 + \gamma_2 B + \gamma_2^2 B^2 + \gamma_2^3 B^3 + \dots\end{aligned}$$

Then

$$\begin{aligned}
X_t &= \frac{1}{\gamma_2 - \gamma_1} (\gamma_2(1 + \gamma_2 B + \gamma_2^2 B^2 + \dots) - \gamma_1(1 + \gamma_1 B + \gamma_1^2 B^2 + \dots)) Z_t \\
&= \frac{1}{\gamma_2 - \gamma_1} (\gamma_2 + \gamma_2^2 B + \gamma_2^3 B^2 + \dots - \gamma_1 - \gamma_1^2 B - \gamma_1^3 B^2 - \dots) Z_t \\
&= \frac{1}{\gamma_2 - \gamma_1} (\gamma_2 - \gamma_1 + (\gamma_2^2 - \gamma_1^2)B + (\gamma_2^3 - \gamma_1^3)B^2 + \dots) Z_t \\
&= \frac{1}{\gamma_2 - \gamma_1} \sum_{k=0}^{\infty} (\gamma_2^{k+1} - \gamma_1^{k+1}) Z_{t-k}.
\end{aligned}$$

(c) Derive the autocovariance function of  $\{X_t\}$ .

**SOLUTION:**

Assuming  $h > 0$  we write

$$X_{t+h} = \frac{1}{\gamma_2 - \gamma_1} \sum_{j=0}^{\infty} (\gamma_2^{j+1} - \gamma_1^{j+1}) Z_{t+h-j}.$$

Since the process  $\{X_t\}$  has mean zero we deduce that

$$\begin{aligned}
\gamma(h) &= \mathbb{E}(X_t X_{t+h}) \\
&= \mathbb{E} \left[ \left( \frac{1}{\gamma_2 - \gamma_1} \sum_{k=0}^{\infty} (\gamma_2^{k+1} - \gamma_1^{k+1}) Z_{t-k} \right) \left( \frac{1}{\gamma_2 - \gamma_1} \sum_{j=0}^{\infty} (\gamma_2^{j+1} - \gamma_1^{j+1}) Z_{t+h-j} \right) \right] \\
&= \frac{1}{(\gamma_2 - \gamma_1)^2} \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} (\gamma_2^{k+1} - \gamma_1^{k+1})(\gamma_2^{j+1} - \gamma_1^{j+1}) \mathbb{E}(Z_{t-k} Z_{t+h-j})
\end{aligned}$$

The only non-zero terms in the above sum occur where  $k = -h + j \implies j = k + h$ , then

$$\begin{aligned}
\gamma(h) &= \frac{\sigma_z^2}{(\gamma_2 - \gamma_1)^2} \sum_{k=0}^{\infty} (\gamma_2^{k+1} - \gamma_1^{k+1})(\gamma_2^{k+h+1} - \gamma_1^{k+h+1}) \\
&= \frac{\sigma_z^2}{(\gamma_2 - \gamma_1)^2} \left\{ \sum_{k=0}^{\infty} \gamma_2^{2k+h+2} - \sum_{k=0}^{\infty} \gamma_1^{k+1} \gamma_2^{k+h+1} - \sum_{k=0}^{\infty} \gamma_1^{k+h+1} \gamma_2^{k+1} + \sum_{k=0}^{\infty} \gamma_1^{2k+h+2} \right\}
\end{aligned}$$

Each of the above sums is the sum to infinity of a geometric progression with  $|r| < 1$ . Hence

$$\gamma(h) = \frac{\sigma_z^2}{(\gamma_2 - \gamma_1)^2} \left[ \frac{\gamma_2^{h+2}}{1 - \gamma_2^2} - \frac{\gamma_1 \gamma_2^{h+1}}{1 - \gamma_1 \gamma_2} - \frac{\gamma_2 \gamma_1^{h+1}}{1 - \gamma_1 \gamma_2} + \frac{\gamma_1^{h+2}}{1 - \gamma_1^2} \right].$$

3. For each of the following ARMA processes, determine: (i) whether or not the process is stationary; (ii) whether or not the process is invertible. In each case,  $\{Z_t\}$  is a white noise process with variance  $\sigma_z^2$ .

(a)  $X_t + 0.2X_{t-1} - 0.48X_{t-2} = Z_t$

**SOLUTION:**

Since this is an  $AR(2)$  process written

$$(1 + 0.2B - 0.48B^2)X_t = Z_t$$

For stationarity, we check whether or not roots of

$$1 + 0.2z - 0.48z^2 = 0$$

lie outside the unit circle.

$$\begin{aligned} 1 + 0.2z - 0.48z^2 &= 0 \\ \implies z &= \frac{-0.2 \pm \sqrt{0.04 + 1.92}}{-0.96} \\ z &= -1.25 \text{ or } 1.67. \end{aligned}$$

Each root lies outside the unit circle so the process is (weakly) stationary. Note: we could also use the stationarity conditions derived during Lecture 4 (slides 37–39).

Since the process is an autoregressive process then, by definition, the process is invertible.

(b)  $X_t + 1.9X_{t-1} + 0.88X_{t-2} = Z_t + 0.2Z_{t-1} + 0.7Z_{t-2}$

**SOLUTION:**

This is an  $ARMA(2, 2)$  process written

$$\phi(B)X_t = \theta(B)Z_t$$

with

$$\begin{aligned} \phi(B) &= 1 + 1.9B + 0.88B^2 \\ \theta(B) &= 1 + 0.2B + 0.7B^2 \end{aligned}$$

For stationarity, we check whether or not roots of  $\phi(z) = 0$  lie outside the unit circle.

$$\begin{aligned} 1 + 1.9z + 0.88z^2 &= 0 \\ \implies z &= \frac{-1.9 \pm \sqrt{1.9^2 - 4 \cdot 0.88}}{2 \cdot 0.88} \\ &= \frac{-1.9 \pm \sqrt{0.09}}{1.76} \\ &= -1.25 \text{ or } -0.9090. \end{aligned}$$

One root lies outside the unit circle so the process is not stationary.

For invertibility we check whether or not roots of  $\theta(z) = 0$  lie outside the unit circle.

$$\begin{aligned} 1 + 0.2z + 0.7z^2 &= 0 \\ \implies z &= \frac{-0.2 \pm \sqrt{0.04 - 2.8}}{1.4} \\ &= \frac{-0.2 \pm \sqrt{-2.76}}{1.4} \end{aligned}$$

Clearly, we have a pair of complex roots, with modulus

$$\sqrt{\frac{0.2^2}{1.4^2} + \frac{2.76}{1.4^2}} = 1.428 > 1$$

so the roots lie outside the unit circle and the process is invertible.

(c)  $X_t + 0.6X_{t-1} = Z_t + 1.2Z_{t-1}$

**SOLUTION:**

This is an ARMA(1,1) process where

$$\phi(B)X_t = \theta(B)Z_t$$

with

$$\begin{aligned} \phi(B) &= 1 + 0.6B \\ \theta(B) &= 1 + 1.2B \end{aligned}$$

The root of  $\phi(z) = 0$  is  $-1/0.6$  which has magnitude greater than 1 (i.e. the root lies outside the unit circle). Hence the process is (weakly) stationary.

The root of  $\theta(z) = 0$  is  $-1/1.2$  which has magnitude less than 1 (i.e. the root lies inside the unit circle). Hence the process is not invertible.

(d)  $X_t + 1.8X_{t-1} + 0.81X_{t-2} = Z_t$

**SOLUTION:**

This is an  $AR(2)$  process so, by definition, it is invertible. Now, as a different solution, let's explore stationarity using the conditions derived during Lecture 4. Recall that if

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + Z_t$$

is an  $AR(2)$  process then the process is (weakly) stationary if

$$\phi_1 + \phi_2 < 1; \quad \phi_2 - \phi_1 < 1 \quad \text{and} \quad |\phi_2| < 1.$$

Here we have  $\phi_1 = -1.8, \phi_2 = -0.81$  and

$$\phi_1 + \phi_2 = -1.8 - 0.81 < 1;$$

$$\phi_2 - \phi_1 = -0.81 - 1.8 < 1;$$

$$|\phi_2| = 0.81 < 1.$$

Hence the process is (weakly) stationary.

(e)  $X_t + 1.6X_{t-1} = Z_t - 0.3Z_{t-1}$

**SOLUTION:**

This is an  $ARMA(1, 1)$  process where

$$\phi(B)X_t = \theta(B)Z_t$$

with

$$\phi(B) = 1 + 1.6B$$

$$\theta(B) = 1 - 0.3B$$

The root of  $\phi(B) = 0$  is  $-1/1.6$  which has magnitude less than 1 (i.e. the root lies inside the unit circle). Hence the process is not stationary.

The root of  $\theta(B) = 0$  is  $1/0.3$  which has magnitude greater than 1 (i.e. the root lies outside the unit circle). Hence the process is invertible.

4. Suppose that the time series  $\{Y_t\}$  is defined

$$Y_t = X_t + W_t$$

where  $\{W_t\}$  is a white noise process with variance  $\sigma_w^2$  and  $\{X_t\}$  is an  $AR(1)$  process such that

$$X_t = \phi X_{t-1} + Z_t$$

where  $|\phi| < 1$ ,  $\{Z_t\}$  is a white noise process with  $\text{Var}(Z_t) = \sigma_z^2$  and  $\mathbb{E}(W_s Z_t) = 0$  for all  $s$  and  $t$ .

(a) Show that the process  $\{Y_t\}$  is stationary and determine its autocovariance function.

**SOLUTION:**

Firstly, note that the process  $\{X_t\}$  is (weakly) stationary because  $|\phi| < 1$ . Writing  $\{X_t\}$  in  $MA(\infty)$  form we see that

$$X_t = \sum_{j=0}^{\infty} \phi^j Z_{t-j}.$$

We deduce that  $\mathbb{E}(X_t) = 0$ .

Consider  $\mathbb{E}(Y_t)$

$$\begin{aligned}\mathbb{E}(Y_t) &= \mathbb{E}(X_t + W_t) \\ &= \mathbb{E}(X_t) + \mathbb{E}(W_t) \\ &= 0\end{aligned}$$

so the mean of  $Y_t$  is constant. Now, let  $\gamma_X(h)$  and  $\gamma_Y(h)$  be the autocovariance functions (at lag  $h$ ) of  $\{X_t\}$  and  $\{Y_t\}$ , respectively. Then

$$\begin{aligned}\gamma_Y(0) &= \text{Var}(Y_t) \\ &= \text{Var}(X_t + W_t) \\ &= \text{Var}(X_t) + \text{Var}(W_t) \quad (\text{since } X_t \text{ and } W_t \text{ are independent}) \\ &= \gamma_X(0) + \sigma_w^2\end{aligned}$$

which is finite and does not depend on  $t$  (because  $\{X_t\}$  is stationary).

Finally, consider  $\gamma_Y(h)$

$$\begin{aligned}\gamma_Y(h) &= \text{Cov}(Y_t, Y_{t+h}) \\ &= \mathbb{E}(Y_t Y_{t+h}) \\ &= \mathbb{E}[(X_t + W_t)(X_{t+h} + W_{t+h})] \\ &= \mathbb{E}(X_t X_{t+h}) + \mathbb{E}(W_t X_{t+h}) + \mathbb{E}(X_t W_{t+h}) + \mathbb{E}(W_t W_{t+h}) \\ &= \mathbb{E}(X_t X_{t+h}) \\ &= \gamma_X(h).\end{aligned}$$

Since  $\gamma_X(h)$  does not depend on  $t$  (because  $\{X_t\}$  is a stationary process) it follows that  $\{Y_t\}$  is also stationary. Moreover, from lectures we know that

$$\gamma_Y(h) = \begin{cases} \frac{\sigma_z^2}{1-\phi^2} + \sigma_w^2 & h = 0; \\ \frac{\sigma_z^2}{1-\phi^2} \phi^{|h|} & |h| \geq 1. \end{cases}$$

(b) Show that the process  $\{U_t\}$ , where  $U_t = Y_t - \phi Y_{t-1}$ , is an  $MA(1)$  process.

**SOLUTION:**

We see that

$$\begin{aligned}
U_t &= Y_t - \phi Y_{t-1} \\
&= (X_t + W_t) - \phi(X_{t-1} + W_{t-1}) \\
&= X_t - \phi X_{t-1} + W_t - \phi W_{t-1} \\
&= Z_t + W_t - \phi W_{t-1}.
\end{aligned}$$

Then

$$\begin{aligned}
\gamma_U(0) &= \text{Var}(U_t) = \text{Var}(Z_t + W_t - \phi W_{t-1}) \\
&= \text{Var}(Z_t) + \text{Var}(W_t) + \phi^2 \text{Var}(W_{t-1}) \\
&= \sigma_z^2 + \sigma_w^2 + \phi^2 \sigma_w^2 \\
&= \sigma_z^2 + \sigma_w^2(1 + \phi^2).
\end{aligned}$$

Also

$$\begin{aligned}
\gamma_U(1) &= \text{Cov}(U_t, U_{t+1}) = \mathbb{E}[(Z_t + W_t - \phi W_{t-1})(Z_{t+1} + W_{t+1} - \phi W_t)] \\
&= -\phi \mathbb{E}(W_t^2) \\
&= -\phi \sigma_w^2
\end{aligned}$$

and  $\gamma_U(h) = 0$  for  $|h| > 1$ . Hence,  $\{U_t\}$  is an  $MA(1)$  process.

- (c) Show that  $\{Y_t\}$  is an  $ARMA(1,1)$  process and determine the parameters of this process in terms of  $\phi$ ,  $\sigma_w^2$  and  $\sigma_z^2$ .

**SOLUTION:**

From (b) we know that

$$Y_t - \phi Y_t = U_t$$

and  $U_t$  is an  $MA(1)$  process. Hence

$$U_t = V_t + \theta V_{t-1}$$

for some  $\theta$  where  $\{V_t\}$  is a white noise process with variance  $\sigma_v^2$ . Then

$$Y_t - \phi Y_t = V_t + \theta V_{t-1}$$

and it follows that  $\{Y_t\}$  is an  $ARMA(1,1)$  process. Clearly the AR parameter is  $\phi$  but we need to find the other parameters  $\theta$  and  $\sigma_v^2$  in terms of  $\sigma_w^2$ ,  $\sigma_z^2$  and  $\phi$ . We note that

$$\begin{aligned}
\text{Var}(U_t) &= \text{Var}(V_t + \theta V_{t-1}) \\
&= \text{Var}(V_t) + \theta^2 \text{Var}(V_{t-1}) \\
&= \sigma_v^2(1 + \theta^2)
\end{aligned}$$

From (b),  $\text{Var}(U_t) = \sigma_z^2 + \sigma_w^2(1 + \phi^2)$  which implies that

$$\sigma_v^2(1 + \theta^2) = \sigma_z^2 + \sigma_w^2(1 + \phi^2). \quad (1)$$

In addition, because  $\{U_t\}$  is an  $MA(1)$  process

$$\gamma_U(1) = \sigma_v^2 \theta$$

Also from (b)

$$\gamma_U(1) = -\phi \sigma_w^2$$

and hence

$$\begin{aligned} \sigma_v^2 \theta &= -\phi \sigma_w^2 \\ \implies \theta &= -\frac{\phi \sigma_w^2}{\sigma_v^2}. \end{aligned}$$

Substitution of the above into (1) yields

$$\begin{aligned} \sigma_v^2 \left(1 + \frac{\phi^2 \sigma_w^4}{\sigma_v^4}\right) &= \sigma_z^2 + \sigma_w^2(1 + \phi^2) \\ \implies \sigma_v^2 \left(\frac{\sigma_v^4 + \phi^2 \sigma_w^4}{\sigma_v^4}\right) &= \sigma_z^2 + \sigma_w^2(1 + \phi^2) \\ \implies \frac{1}{\sigma_v^2} (\sigma_v^4 + \phi^2 \sigma_w^4) &= \sigma_z^2 + \sigma_w^2(1 + \phi^2) \\ \implies \sigma_v^4 + \phi^2 \sigma_w^4 &= \sigma_v^2 (\sigma_z^2 + \sigma_w^2(1 + \phi^2)) \\ \implies \sigma_v^4 - (\sigma_z^2 + \sigma_w^2(1 + \phi^2)) \sigma_v^2 + \phi^2 \sigma_w^4 &= 0 \end{aligned}$$

The above is a quadratic equation for  $\sigma_v^2$  and the solutions are

$$\sigma_v^2 = \frac{(\sigma_z^2 + \sigma_w^2(1 + \phi^2)) \pm \sqrt{(\sigma_z^2 + \sigma_w^2(1 + \phi^2))^2 - 4\phi^2 \sigma_w^4}}{2}.$$

We'll choose the larger root to ensure that the  $MA(1)$  process is invertible. Then

$$\sigma_v^2 = \frac{(\sigma_z^2 + \sigma_w^2(1 + \phi^2)) + \sqrt{(\sigma_z^2 + \sigma_w^2(1 + \phi^2))^2 - 4\phi^2 \sigma_w^4}}{2}$$

and

$$\begin{aligned} \theta &= -\frac{\phi \sigma_w^2}{\sigma_v^2} \\ &= \frac{-2\phi \sigma_w^2}{(\sigma_z^2 + \sigma_w^2(1 + \phi^2)) + \sqrt{(\sigma_z^2 + \sigma_w^2(1 + \phi^2))^2 - 4\phi^2 \sigma_w^4}}. \end{aligned}$$

5. The process  $\{X_t\}$  is the AR(2) process, written

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + Z_t,$$

where  $\{Z_t\}$  is a white noise process with variance  $\sigma_z^2$ . Suppose that  $\rho(1)$  and  $\rho(2)$  denote the autocorrelations for this process at lags 1 and 2, respectively.

Prove that the following conditions are necessary and sufficient for the AR(2) process to be weakly stationary:

$$\begin{aligned} -1 &< \rho(1) < 1; \\ -1 &< \rho(2) < 1; \\ \rho(1)^2 &< \frac{1}{2}[\rho(2) + 1]. \end{aligned}$$

### SOLUTION

Recall that if

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + Z_t$$

is an AR(2) process then the process is (weakly) stationary if

$$\phi_1 + \phi_2 < 1; \quad \phi_2 - \phi_1 < 1 \quad \text{and} \quad |\phi_2| < 1.$$

Let us assume that  $\{X_t\}$  is (weakly) stationary and so the above inequalities hold.

From Lecture 4

$$\rho(1) = \frac{\phi_1}{1 - \phi_2} \quad \text{and} \quad \rho(2) = \phi_2 + \frac{\phi_1^2}{1 - \phi_2}$$

Since  $|\phi_2| < 1$  then it follows that  $1 - \phi_2 > 0$  hence

$$|\rho(1)| = \frac{|\phi_1|}{1 - \phi_2}$$

Since  $\phi_1 + \phi_2 < 1$  then  $\phi_1 < 1 - \phi_2$  and, hence,

$$|\rho(1)| = \frac{|\phi_1|}{1 - \phi_2} < \frac{|1 - \phi_2|}{|1 - \phi_2|} = 1$$

So  $|\rho(1)| < 1$ .

Also

$$\begin{aligned} \rho(2) &= \phi_2 + \frac{\phi_1^2}{1 - \phi_2} < \phi_2 + \frac{(1 - \phi_2)^2}{1 - \phi_2} \\ &\implies \rho(2) < \phi_2 + 1 - \phi_2 = 1. \end{aligned} \tag{1}$$

We note that

$$\rho(2) = \phi_2 + \frac{\phi_1^2}{1 - \phi_2} = \phi_2 + (1 - \phi_2)\rho(1)^2.$$

Then

$$\begin{aligned}\rho(2) &= \rho(1)^2 + \phi_2(1 - \rho(1)^2) \\ &> \rho(1)^2 - (1 - \rho(1)^2) \quad (\text{because } \phi_2 > -1). \\ \implies \rho(2) &> 2\rho(1)^2 - 1\end{aligned}$$

and it follows that

$$\rho(1)^2 < \frac{1}{2}[\rho(2) + 1].$$

Finally, because  $\rho(1)^2 \geq 0$  we see that

$$\frac{1}{2}[\rho(2) + 1] > 0 \implies \rho(2) > -1.$$

Combining the above with (1), we see that  $-1 < \rho(2) < 1$ .

Hence, for a stationary  $AR(2)$  process

$$\begin{aligned}-1 &< \rho(1) < 1; \\ -1 &< \rho(2) < 1; \\ \rho(1)^2 &< \frac{1}{2}[\rho(2) + 1].\end{aligned}$$

6. The process  $\{X_t\}$  is an  $AR(2)$  process where

$$X_t = \frac{2}{3}X_{t-1} - \frac{2}{9}X_{t-2} + Z_t.$$

(a) Write down the Yule-Walker equations for this process.

**SOLUTION:**

The Yule-Walker equations are

$$\rho(h) - \frac{2}{3}\rho(h-1) + \frac{2}{9}\rho(h-2) = 0 \quad \text{for } h \geq 2.$$

(b) By solving the Yule-Walker equations, determine the autocorrelation function of  $\{X_t\}$ .

**SOLUTION:**

The auxiliary equation is

$$\begin{aligned}
\lambda^2 - \frac{2}{3}\lambda + \frac{2}{9} &= 0 \\
\Rightarrow \left(\lambda - \frac{1}{3}\right)^2 - \frac{1}{9} + \frac{2}{9} &= 0 \\
\Rightarrow \left(\lambda - \frac{1}{3}\right)^2 &= -\frac{1}{9} \\
\Rightarrow \lambda &= \frac{1}{3} \pm \frac{i}{3} \\
&= \frac{\sqrt{2}}{3} e^{\pm i\frac{\pi}{4}}.
\end{aligned}$$

The general solution has the form

$$\rho(h) = C_1 \left(\frac{\sqrt{2}}{3}\right)^{|h|} e^{\frac{i\pi|h|}{4}} + C_2 \left(\frac{\sqrt{2}}{3}\right)^{|h|} e^{\frac{-i\pi|h|}{4}}.$$

Since  $\rho(0) = 1$

$$C_1 + C_2 = 1$$

Also

$$\begin{aligned}
\rho(1) - \frac{2}{3}\rho(0) + \frac{2}{9}\rho(-1) &= 0 \\
\Rightarrow \frac{11}{9}\rho(1) &= \frac{2}{3} \\
\Rightarrow \rho(1) &= \frac{6}{11}.
\end{aligned}$$

Then

$$\begin{aligned}
C_1 \left(\frac{\sqrt{2}}{3}\right) e^{\frac{i\pi}{4}} + C_2 \left(\frac{\sqrt{2}}{3}\right) e^{\frac{-i\pi}{4}} &= \frac{6}{11} \\
C_1 \left(\frac{\sqrt{2}}{3}\right) e^{\frac{i\pi}{4}} + (1 - C_1) \left(\frac{\sqrt{2}}{3}\right) e^{\frac{-i\pi}{4}} &= \frac{6}{11} \\
\Rightarrow \frac{2\sqrt{2}}{3} C_1 i \sin\left(\frac{\pi}{4}\right) &= \frac{6}{11} - \left(\frac{\sqrt{2}}{3}\right) e^{\frac{-i\pi}{4}} \\
\Rightarrow C_1 \frac{2i}{3} &= \frac{6}{11} - \left(\frac{1}{3} - \frac{i}{3}\right) \\
&= \frac{1}{3} \left(\frac{7}{11} + i\right) \\
\Rightarrow C_1 &= \frac{1}{2i} \left(\frac{7}{11} + i\right) \\
&= \frac{1}{2} - \frac{7i}{22}
\end{aligned}$$

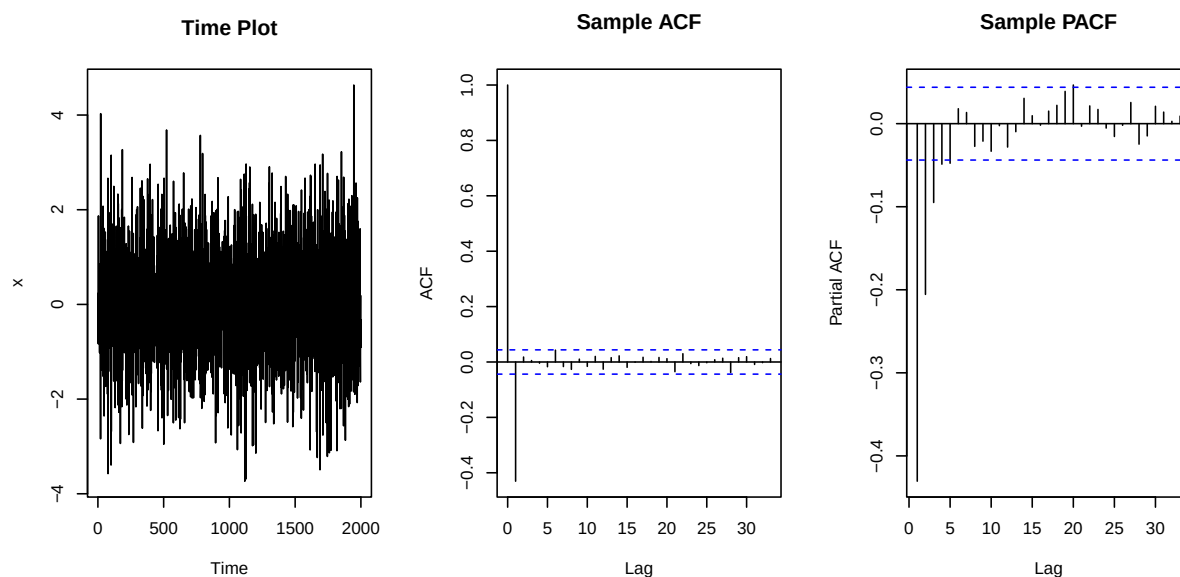
Hence

$$C_2 = \frac{1}{2} + \frac{7i}{22}.$$

Then

$$\begin{aligned}\rho(h) &= C_1 \left( \frac{\sqrt{2}}{3} \right)^{|h|} e^{\frac{i\pi|h|}{4}} + C_2 \left( \frac{\sqrt{2}}{3} \right)^{|h|} e^{\frac{-i\pi|h|}{4}} \\ &= \left( \frac{1}{2} - \frac{7i}{22} \right) \left( \frac{\sqrt{2}}{3} \right)^{|h|} \left( \cos \left( \frac{\pi|h|}{4} \right) + i \sin \left( \frac{\pi|h|}{4} \right) \right) \\ &\quad + \left( \frac{1}{2} + \frac{7i}{22} \right) \left( \frac{\sqrt{2}}{3} \right)^{|h|} \left( \cos \left( \frac{\pi|h|}{4} \right) - i \sin \left( \frac{\pi|h|}{4} \right) \right) \\ &= \left( \frac{\sqrt{2}}{3} \right)^{|h|} \left( \cos \left( \frac{\pi|h|}{4} \right) + \frac{7}{11} \sin \left( \frac{\pi|h|}{4} \right) \right).\end{aligned}$$

7. A statistician would like to fit an ARMA model to a time series, where 2000 sequential observations have been observed. The observations provided sample autocorrelations and sample partial autocorrelations, shown in the plots below, together with a time plot.



- (a) Using these plots, identify the type of ARMA model that might fit the data. Justify your answer.

**SOLUTION:**

Looking at the plots, the sample ACF appears to ‘cut off’ after lag 1 - this may suggest an MA component to the model with order 1. Conversely, the sample PACF does not appear to cut off after particular lag which suggests that there is not an AR component present. As such, we might consider fitting an  $MA(1)$  model.

- (b) Determine possible parameter values for the process chosen in (a).

**SOLUTION:**

Suppose that we want to fit an  $MA(1)$  model to these data. The two parameters of interest are  $\theta$  and  $\sigma_z^2$  where

$$X_t = Z_t + \theta Z_{t-1}$$

with  $\text{Var}(Z_t) = \sigma^2$ . For the  $MA(1)$  process, we know that the theoretical ACF at lag 1 is

$$\rho(1) = \frac{\theta}{1 + \theta^2}.$$

Looking at the sample ACF plot,  $\rho(1)$  is approximately -0.4. Therefore set

$$\begin{aligned}\frac{\theta}{1 + \theta^2} &= -0.4 \\ \theta &= -0.4(1 + \theta^2) \\ \theta^2 + \frac{5}{2}\theta + 1 &= 0 \\ \left(\theta + \frac{5}{4}\right)^2 - \frac{9}{16} &= 0 \\ \theta + \frac{5}{4} &= \pm \frac{3}{4} \\ \theta &= -\frac{5}{4} \pm \frac{3}{4}\end{aligned}$$

Hence  $\theta = -\frac{1}{2}$  or  $\theta = -2$ . We choose  $\theta = -\frac{1}{2}$  so that the process is invertible.

To estimate  $\sigma_z^2$  it might be best calculate the sample variance of the sampled values,  $s^2$ , and equate this to

$$s^2 = \sigma_z^2(1 + \theta^2).$$

substituting in  $\theta = -0.5$  and then solving for  $\sigma_z^2$ .

Alternatively, as an approximate ‘guess’ for  $\sigma_z^2$ , note from the time plot that most sampled values lie between  $-\frac{5}{2}$  and  $+\frac{5}{2}$ . If we assume that the interval  $(-\frac{5}{2}, \frac{5}{2})$  represents  $\pm$  two standard deviations of the sampled values, then set

$$2s = \frac{5}{2} \implies s = \frac{5}{4}$$

where  $s$  is the standard deviation of the sampled values. Then

$$\begin{aligned}s^2 &= \hat{\sigma}_z^2(1 + (-0.5)^2) \quad (\text{setting } \theta = -0.5) \\ \frac{5^2}{4^2} &= \frac{5}{4}\hat{\sigma}_z^2 \\ \implies \hat{\sigma}_z^2 &= \frac{5}{4} \\ \implies \hat{\sigma}_z &\approx 1.118.\end{aligned}$$